Based on the configuration of your Firebase project PlantFlow (plantflow-os), here is a high-level security and architectural audit of your current setup:

1. Billing & Multi-Database Architecture

Your project is on the paid Blaze plan, which extends your capabilities beyond the no-cost tier. This is a great choice as it enables your multi-database Firestore configuration. Running multiple databases in europe-west2 is an excellent strategy for isolating development stages or separating different segments of your application.

2. Firestore Security Rules

- Database ai-studio-e467...: Currently configured with a "deny-all" security rule. This is highly secure and is the correct approach if this database is exclusively accessed by privileged server-side environments (such as Cloud Functions) rather than client-side SDKs.

- Database ai-studio-a0ff...: This database allows read and write access to any authenticated user. While this prevents unauthenticated public access, it still allows any logged-in user to read or modify any document in your database. This represents a security vulnerability for production; you should refine these rules to ensure users can only access data they own.

3. Authentication & Identity Platform

- Identity Platform: You have successfully upgraded to Identity Platform, which unlocks enterprise-grade authentication features.

- MFA (Multi-Factor Authentication): This is currently disabled. For production environments, enabling MFA is highly recommended to protect user accounts.

- Sign-In Methods: Only Google Sign-In is enabled. Since your apps are TypeScript/JavaScript Web apps, if you want to support users without Google accounts, you should consider enabling other providers like Email/Password.

- Authorized Domains: These are properly locked down to your default Firebase Hosting domains and localhost for development.

To explore these topics in more detail, you could ask:

1 How can I configure my Firestore rules on the second database to restrict users to only their own documents?

2 What are the steps to enable and enforce Multi-Factor Authentication (MFA) using Identity Platform in my TypeScript apps?

3 How do I programmatically configure my web apps to connect to multiple Firestore databases?

- Audit logging for Firebase Management
  - # Audit logging for Firebase Management This document describes audit logging for Firebase Management. Google Cloud services generate audit logs that record administrative and access activities within your Google Cloud resources. For more information about Cloud Audit Logs, see the following: ∗ [Types of audit logs](https://cloud.google.com/logging/docs/audit#types) ∗ [Audit log entry structure](https://cloud.google.com/logging/docs/audit#audit_log_entry_structure) ∗ [Storing and routing audit logs](https://cloud.google.com/logging/docs/audit#storing_and_routing_audit_logs) ∗ [Cloud Logging pricing summary](https://cloud.google.com/stackdriver/pricing#logs-pricing-summary) ∗ [Enable Data Access audit logs](https://cloud.google.com/logging/docs/audit/configure-data-access) ## Service name Firebase Management audit logs use the service name firebase.googleapis.com . Filter for this service: protoPayload . serviceName = "firebase.googleapis.com" # Audit logging for Firebase Management ## Methods by permission type Each IAM permission has a type property, whose value is an enum that can be one of four values: ADMIN_READ , ADMIN_WRITE , DATA_READ , or DATA_WRITE . When you call a method, Firebase Management generates an audit log whose category is dependent on the type property of the permission required to perform the method. Methods that require an IAM permission with the type property value of DATA_READ , DATA_WRITE , or ADMIN_READ generate [Data Access](https://cloud.google.com/logging/docs/audit#data-access)audit logs. Methods that require an IAM permission with the type property value of ADMIN_WRITE generate [Admin Activity](https://cloud.google.com/logging/docs/audit#admin-activity)audit logs. API methods in the following list that are marked with (LRO) are long-running operations (LROs). These methods usually generate two audit log entries: one when the operation starts and another when it ends. For more information see [Audit logs for long-running operations](https://cloud.google.com/logging/docs/audit/understanding-audit-logs#lro). | Permission type | Methods | |-|-| | ADMIN_ READ | google. firebase. service. v1beta1. Android App Service. Get Android App google. firebase. service. v1beta1. Android App Service. Get Android App Config google. firebase. service. v1beta1. Android App Service. List Android Apps google. firebase. service. v1beta1. Android App Service. List Sha Certificates google. firebase. service. v1beta1. Firebase Project Service. Get Admin Sdk Config google. firebase. service. v1beta1. Firebase Project Service. Get Firebase Project google. firebase. service. v1beta1. Firebase Project Service. List Available Locations google. firebase. service.

v1beta1. Firebase Project Service. Search Firebase Apps google. firebase. service. v1beta1. Ios App Service. Get Ios App google. firebase. service. v1beta1. Ios App Service. Get Ios App Config google. firebase. service. v1beta1. Ios App Service. List Ios Apps google. firebase. service. v1beta1. Web App Service. Get Web App google. firebase. service. v1beta1. Web App Service. Get Web App Config google. firebase. service. v1beta1. Web App Service. List Web Apps | | ADMIN_ WRITE | google. firebase. service. v1beta1. Android App Service. Create Android App (LRO) google. firebase. service. v1beta1. Android App Service. Create Sha Certificate google. firebase. service. v1beta1. Android App Service. Delete Sha Certificate google. firebase. service. v1beta1. Android App Service. Remove Android App google. firebase. service. v1beta1. Android App Service. Undelete Android App google. firebase. service. v1beta1. Android App Service. Update Android App google. firebase. service. v1beta1. Firebase Project Service. Add Firebase (LRO) google. firebase. service. v1beta1. Firebase Project Service. Finalize Default Location google. firebase. service. v1beta1. Firebase Project Service. Update Firebase Project google. firebase. service. v1beta1. Ios App Service. Create Ios App (LRO) google. firebase. service. v1beta1. Ios App Service. Remove Ios App google. firebase. service. v1beta1. Ios App Service. Undelete Ios App google. firebase. service. v1beta1. Ios App Service. Update Ios App google. firebase. service. v1beta1. Web App Service. Create Web App (LRO) google. firebase. service. v1beta1. Web App Service. Remove Web App google. firebase. service. v1beta1. Web App Service. Undelete Web App google. firebase. service. v1beta1. Web App Service. Update Web App | # Audit logging for Firebase Management ## API interface audit logs For information about how and which permissions are evaluated for each method, see the Cloud Identity and Access Management documentation for Firebase Management. ### google. firebase. service. v1beta1. Android App Service The following audit logs are associated with methods belonging to google.firebase.service.v1beta1.AndroidAppService .Create Android App ∗ Method : google.firebase.service.v1beta1.AndroidAppService.CreateAndroidApp ∗ Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) ∗ Permissions : ∗ firebase.clients.create - ADMIN_WRITE ∗ Method is a long-running or streaming operation : [Long-running operation](https://cloud.google.com/logging/docs/audit/understanding-audit-logs#lro) ∗ Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.AndroidAppService.CreateAndroidApp" Create Sha Certificate ∗ Method : google.firebase.service.v1beta1.AndroidAppService.CreateShaCertificate ∗ Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) ∗ Permissions : ∗ firebase.clients.update - ADMIN_WRITE ∗ Method is a long-running or streaming operation : No. ∗ Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.AndroidAppService.CreateShaCertificate" Delete Sha Certificate ∗ Method : google.firebase.service.v1beta1.AndroidAppService.DeleteShaCertificate ∗ Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) ∗ Permissions : ∗ firebase.clients.update - ADMIN_WRITE ∗ Method is a long-running or

streaming operation : No. ∗ Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.AndroidAppService.DeleteShaCertificate" Get Android App ∗ Method : google.firebase.service.v1beta1.AndroidAppService.GetAndroidApp ∗ Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) ∗ Permissions : ∗ firebase.clients.get - ADMIN_READ ∗ Method is a long-running or streaming operation : No. ∗ Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.AndroidAppService.GetAndroidApp" Get Android App Config ∗ Method : google.firebase.service.v1beta1.AndroidAppService.GetAndroidAppConfig ∗ Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) ∗ Permissions : ∗ firebase.clients.get - ADMIN_READ ∗ Method is a long-running or streaming operation : No. ∗ Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.AndroidAppService.GetAndroidAppConfig" List Android Apps ∗ Method : google.firebase.service.v1beta1.AndroidAppService.ListAndroidApps ∗ Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) ∗ Permissions : ∗ firebase.clients.list - ADMIN_READ ∗ Method is a long-running or streaming operation : No. ∗ Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.AndroidAppService.ListAndroidApps" List Sha Certificates ∗ Method : google.firebase.service.v1beta1.AndroidAppService.ListShaCertificates ∗ Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) ∗ Permissions : ∗ firebase.clients.get - ADMIN_READ ∗ Method is a long-running or streaming operation : No. ∗ Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.AndroidAppService.ListShaCertificates" Remove Android App ∗ Method : google.firebase.service.v1beta1.AndroidAppService.RemoveAndroidApp ∗ Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) ∗ Permissions : ∗ firebase.clients.delete - ADMIN_WRITE ∗ Method is a long-running or streaming operation : No. ∗ Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.AndroidAppService.RemoveAndroidApp" Undelete Android App ∗ Method : google.firebase.service.v1beta1.AndroidAppService.UndeleteAndroidApp ∗ Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) ∗ Permissions : ∗ firebase.clients.undelete - ADMIN_WRITE ∗ Method is a long-running or streaming operation : No. ∗ Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.AndroidAppService.UndeleteAndroidApp" Update Android App ∗ Method : google.firebase.service.v1beta1.AndroidAppService.UpdateAndroidApp ∗ Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) ∗ Permissions : ∗ firebase.clients.update - ADMIN_WRITE ∗ Method is a long-running or streaming operation : No. ∗ Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.AndroidAppService.UpdateAndroidApp" ### google. firebase. service. v1beta1. Firebase Analytics Service The following audit logs are associated with methods belonging to

google.firebase.service.v1beta1.FirebaseAnalyticsService .Add Google Analytics * Method : google.firebase.service.v1beta1.FirebaseAnalyticsService.AddGoogleAnalytics * Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) * Permissions : * Method is a long-running or streaming operation : [Long-running operation](https://cloud.google.com/logging/docs/audit/understanding-audit-logs#lro) * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.FirebaseAnalyticsService.AddGoogleAnalytics" # Audit logging for Firebase Management ## API interface audit logs ### google. firebase. service. v1beta1. Firebase Project Service The following audit logs are associated with methods belonging to google.firebase.service.v1beta1.FirebaseProjectService .Add Firebase * Method : google.firebase.service.v1beta1.FirebaseProjectService.AddFirebase * Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) * Permissions : * firebase.projects.update - ADMIN_WRITE * Method is a long-running or streaming operation : [Long-running operation](https://cloud.google.com/logging/docs/audit/understanding-audit-logs#lro) * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.FirebaseProjectService.AddFirebase" Finalize Default Location * Method : google.firebase.service.v1beta1.FirebaseProjectService.FinalizeDefaultLocation * Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) * Permissions : * firebase.projects.update - ADMIN_WRITE * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.FirebaseProjectService.FinalizeDefaultLocation" Get Admin Sdk Config * Method : google.firebase.service.v1beta1.FirebaseProjectService.GetAdminSdkConfig * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebase.projects.get - ADMIN_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.FirebaseProjectService.GetAdminSdkConfig" Get Firebase Project * Method : google.firebase.service.v1beta1.FirebaseProjectService.GetFirebaseProject * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebase.projects.get - ADMIN_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.FirebaseProjectService.GetFirebaseProject" List Available Locations * Method : google.firebase.service.v1beta1.FirebaseProjectService.ListAvailableLocations * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebase.projects.get - ADMIN_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.FirebaseProjectService.ListAvailableLocations" Search Firebase Apps * Method : google.firebase.service.v1beta1.FirebaseProjectService.SearchFirebaseApps * Audit log type : [Data access](https://cloud.google.com/

logging/docs/audit#data-access) * Permissions : * firebase.clients.list - ADMIN_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.FirebaseProjectService.SearchFirebaseApps" Update Firebase Project * Method : google.firebase.service.v1beta1.FirebaseProjectService.UpdateFirebaseProject * Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) * Permissions : * firebase.projects.update - ADMIN_WRITE * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.FirebaseProjectService.UpdateFirebaseProject" # Audit logging for Firebase Management ## API interface audit logs ### google. firebase. service. v1beta1. Ios App Service The following audit logs are associated with methods belonging to google.firebase.service.v1beta1.IosAppService .Create Ios App * Method : google.firebase.service.v1beta1.IosAppService.CreateIosApp * Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) * Permissions : * firebase.clients.create - ADMIN_WRITE * Method is a long-running or streaming operation : [Long-running operation](https://cloud.google.com/logging/docs/audit/understanding-audit-logs#lro) * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.IosAppService.CreateIosApp" Get Ios App * Method : google.firebase.service.v1beta1.IosAppService.GetIosApp * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebase.clients.get - ADMIN_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.IosAppService.GetIosApp" Get Ios App Config * Method : google.firebase.service.v1beta1.IosAppService.GetIosAppConfig * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebase.clients.get - ADMIN_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.IosAppService.GetIosAppConfig" List Ios Apps * Method : google.firebase.service.v1beta1.IosAppService.ListIosApps * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebase.clients.list - ADMIN_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.IosAppService.ListIosApps" Remove Ios App * Method : google.firebase.service.v1beta1.IosAppService.RemoveIosApp * Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) * Permissions : * firebase.clients.delete - ADMIN_WRITE * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.IosAppService.RemoveIosApp" Undelete Ios App * Method : google.firebase.service.v1beta1.IosAppService.UndeleteIosApp * Audit

log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) * Permissions : * firebase.clients.undelete - ADMIN_WRITE * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.IosAppService.UndeleteIosApp" Update Ios App * Method : google.firebase.service.v1beta1.IosAppService.UpdateIosApp * Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) * Permissions : * firebase.clients.update - ADMIN_WRITE * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.IosAppService.UpdateIosApp" # Audit logging for Firebase Management ## API interface audit logs ### google. firebase. service. v1beta1. Web App Service The following audit logs are associated with methods belonging to google.firebase.service.v1beta1.WebAppService .Create Web App * Method : google.firebase.service.v1beta1.WebAppService.CreateWebApp * Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) * Permissions : * firebase.clients.create - ADMIN_WRITE * Method is a long-running or streaming operation : [Long-running operation](https://cloud.google.com/logging/docs/audit/understanding-audit-logs#lro) * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.WebAppService.CreateWebApp" Get Web App * Method : google.firebase.service.v1beta1.WebAppService.GetWebApp * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebase.clients.get - ADMIN_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.WebAppService.GetWebApp" Get Web App Config * Method : google.firebase.service.v1beta1.WebAppService.GetWebAppConfig * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebase.clients.get - ADMIN_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.WebAppService.GetWebAppConfig" List Web Apps * Method : google.firebase.service.v1beta1.WebAppService.ListWebApps * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebase.clients.list - ADMIN_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.WebAppService.ListWebApps" Remove Web App * Method : google.firebase.service.v1beta1.WebAppService.RemoveWebApp * Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) * Permissions : * firebase.clients.delete - ADMIN_WRITE * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.WebAppService.RemoveWebApp" Undelete Web App * Method : google.firebase.service.v1beta1.WebAppService.UndeleteWebApp * Audit

log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) * Permissions : * firebase.clients.undelete - ADMIN_WRITE * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.WebAppService.UndeleteWebApp" Update Web App * Method : google.firebase.service.v1beta1.WebAppService.UpdateWebApp * Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) * Permissions : * firebase.clients.update - ADMIN_WRITE * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.service.v1beta1.WebAppService.UpdateWebApp" No recommendations at this time. Try [signing in](#)to your Google account.

- Firebase Realtime Database audit logging
  - # Firebase Realtime Database audit logging This document describes audit logging for Firebase Realtime Database. Google Cloud services generate audit logs that record administrative and access activities within your Google Cloud resources. For more information about Cloud Audit Logs, see the following: * [Types of audit logs](https://cloud.google.com/logging/docs/audit#types) * [Audit log entry structure](https://cloud.google.com/logging/docs/audit#audit_log_entry_structure) * [Storing and routing audit logs](https://cloud.google.com/logging/docs/audit#storing_and_routing_audit_logs) * [Cloud Logging pricing summary](https://cloud.google.com/stackdriver/pricing#logs-pricing-summary) * [Enable Data Access audit logs](https://cloud.google.com/logging/docs/audit/configure-data-access) ## Notes Additional information about fields in protoPayload.metadata for DATA_READ and DATA_WRITE operations is available in the [reference documentation](/docs/reference/database/cal/rest/Shared.Types/RealtimeDatabaseAuditMetadata). ## Service name Firebase Realtime Database audit logs use the service name firebasedatabase.googleapis.com . Filter for this service: protoPayload . serviceName = "firebasedatabase.googleapis.com" ## Methods by permission type For google.firebase.database.v1.RealtimeDatabase.Update with Precondition, see associated [Precondition](https://firebase.google.com/docs/reference/database/cal/rest/Shared.Types/RealtimeDatabaseAuditMetadata#RealtimeDatabaseAuditMetadata.Precondition). For google.firebase.database.v1.RealtimeDatabase.Listen and google.firebase.database.v1.RealtimeDatabase.Read see associated [ReadMetadata](https://firebase.google.com/docs/reference/database/cal/rest/Shared.Types/RealtimeDatabaseAuditMetadata?db=markarndt-one-plat-ref-docs-techdebt#realtimedatabaseauditmetadata.querymetadata). For google.firebase.database.v1.RealtimeDatabase.OnDisconnectUpdate and google.firebase.database.v1.RealtimeDatabase.Update see associated [WriteMetadata](https://firebase.google.com/docs/reference/database/cal/rest/Shared.Types/RealtimeDatabaseAuditMetadata#realtimedatabaseauditmetadata.writemetadata). Each IAM permission has a type property, whose value is an enum that can be one of four values: ADMIN_READ , ADMIN_WRITE , DATA_READ , or DATA_WRITE . When you call a method, Firebase

Realtime Database generates an audit log whose category is dependent on the type property of the permission required to perform the method. Methods that require an IAM permission with the type property value of DATA_READ , DATA_WRITE , or ADMIN_READ generate [Data Access](https://cloud.google.com/logging/docs/audit#data-access)audit logs. Methods that require an IAM permission with the type property value of ADMIN_WRITE generate [Admin Activity](https://cloud.google.com/logging/docs/audit#admin-activity)audit logs. API methods in the following list that are marked with (LRO) are long-running operations (LROs). These methods usually generate two audit log entries: one when the operation starts and another when it ends. For more information see [Audit logs for long-running operations](https://cloud.google.com/logging/docs/audit/understanding-audit-logs#lro). | Permission type | Methods | |-|-| | ADMIN_ READ | google. firebase. database. v1beta. Realtime Database Service. Get Database Instance google. firebase. database. v1beta. Realtime Database Service. List Database Instances | | ADMIN_ WRITE | google. firebase. database. v1beta. Realtime Database Service. Create Database Instance google. firebase. database. v1beta. Realtime Database Service. Delete Database Instance google. firebase. database. v1beta. Realtime Database Service. Disable Database Instance google. firebase. database. v1beta. Realtime Database Service. Reenable Database Instance google. firebase. database. v1beta. Realtime Database Service. Undelete Database Instance | | DATA_ READ | google. firebase. database. v1. Realtime Database. Connect google. firebase. database. v1. Realtime Database. Disconnect google. firebase. database. v1. Realtime Database. Listen google. firebase. database. v1. Realtime Database. On Disconnect Cancel google. firebase. database. v1. Realtime Database. Read google. firebase. database. v1. Realtime Database. Unlisten | | DATA_ WRITE | google. firebase. database. v1. Realtime Database. On Disconnect Put google. firebase. database. v1. Realtime Database. On Disconnect Update google. firebase. database. v1. Realtime Database. Run On Disconnect google. firebase. database. v1. Realtime Database. Update google. firebase. database. v1. Realtime Database. Write | # Firebase Realtime Database audit logging ## API interface audit logs For information about how and which permissions are evaluated for each method, see the Cloud Identity and Access Management documentation for Firebase Realtime Database. ### google. firebase. database. v1. Realtime Database The following audit logs are associated with methods belonging to google.firebase.database.v1.RealtimeDatabase .Connect ∗ Method : google.firebase.database.v1.RealtimeDatabase.Connect ∗ Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) ∗ Permissions : ∗ firebasedatabase.data.connect - DATA_READ ∗ Method is a long-running or streaming operation : No. ∗ Filter for this method : protoPayload.methodName="google.firebase.database.v1.RealtimeDatabase.Connect" Disconnect ∗ Method : google.firebase.database.v1.RealtimeDatabase.Disconnect ∗ Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) ∗ Permissions : ∗ firebasedatabase.data.connect - DATA_READ ∗ Method is a long-running or streaming operation : No. ∗ Filter for this method :

protoPayload.methodName="google.firebase.database.v1.RealtimeDatabase.Disconnect" Listen * Method : google.firebase.database.v1.RealtimeDatabase.Listen * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebasedatabase.data.get - DATA_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.database.v1.RealtimeDatabase.Listen" On Disconnect Cancel * Method : google.firebase.database.v1.RealtimeDatabase.OnDisconnectCancel * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebasedatabase.data.cancel - DATA_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.database.v1.RealtimeDatabase.OnDisconnectCancel" On Disconnect Put * Method : google.firebase.database.v1.RealtimeDatabase.OnDisconnectPut * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebasedatabase.data.update - DATA_WRITE * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.database.v1.RealtimeDatabase.OnDisconnectPut" On Disconnect Update * Method : google.firebase.database.v1.RealtimeDatabase.OnDisconnectUpdate * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebasedatabase.data.update - DATA_WRITE * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.database.v1.RealtimeDatabase.OnDisconnectUpdate" Read * Method : google.firebase.database.v1.RealtimeDatabase.Read * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebasedatabase.data.get - DATA_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.database.v1.RealtimeDatabase.Read" Run On Disconnect * Method : google.firebase.database.v1.RealtimeDatabase.RunOnDisconnect * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebasedatabase.data.update - DATA_WRITE * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.database.v1.RealtimeDatabase.RunOnDisconnect" Unlisten * Method : google.firebase.database.v1.RealtimeDatabase.Unlisten * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebasedatabase.data.cancel - DATA_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.database.v1.RealtimeDatabase.Unlisten" Update * Method : google.firebase.database.v1.RealtimeDatabase.Update * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebasedatabase.data.get -

DATA_WRITE * firebasedatabase.data.update - DATA_WRITE * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.database.v1.RealtimeDatabase.Update" Write * Method : google.firebase.database.v1.RealtimeDatabase.Write * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebasedatabase.data.update - DATA_WRITE * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.database.v1.RealtimeDatabase.Write" # Firebase Realtime Database audit logging ## API interface audit logs ### google. firebase. database. v1beta. Realtime Database Service The following audit logs are associated with methods belonging to google.firebase.database.v1beta.RealtimeDatabaseService .Create Database Instance * Method : google.firebase.database.v1beta.RealtimeDatabaseService.CreateDatabaseInstance * Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) * Permissions : * firebasedatabase.instances.create - ADMIN_WRITE * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.database.v1beta.RealtimeDatabaseService.CreateDatabaseInstance" Delete Database Instance * Method : google.firebase.database.v1beta.RealtimeDatabaseService.DeleteDatabaseInstance * Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) * Permissions : * firebasedatabase.instances.delete - ADMIN_WRITE * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.database.v1beta.RealtimeDatabaseService.DeleteDatabaseInstance" Disable Database Instance * Method : google.firebase.database.v1beta.RealtimeDatabaseService.DisableDatabaseInstance * Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) * Permissions : * firebasedatabase.instances.disable - ADMIN_WRITE * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.database.v1beta.RealtimeDatabaseService.DisableDatabaseInstance" Get Database Instance * Method : google.firebase.database.v1beta.RealtimeDatabaseService.GetDatabaseInstance * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebasedatabase.instances.get - ADMIN_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.database.v1beta.RealtimeDatabaseService.GetDatabaseInstance" List Database Instances * Method : google.firebase.database.v1beta.RealtimeDatabaseService.ListDatabaseInstances * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebasedatabase.instances.list - ADMIN_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.firebase.database.v1beta.RealtimeDatabaseService.ListDatabaseInstances" Reenable Database Instance *

Method : google.firebase.database.v1beta.RealtimeDatabaseService.ReenableDatabaseInstance ∗ Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) ∗ Permissions : ∗ firebasedatabase.instances.reenable - ADMIN_WRITE ∗ Method is a long-running or streaming operation : No. ∗ Filter for this method : protoPayload.methodName="google.firebase.database.v1beta.RealtimeDatabaseService.ReenableDatabaseInstance" Undelete Database Instance ∗ Method : google.firebase.database.v1beta.RealtimeDatabaseService.UndeleteDatabaseInstance ∗ Audit log type : [Admin activity](https://cloud.google.com/logging/docs/audit#admin-activity) ∗ Permissions : ∗ firebasedatabase.instances.undelete - ADMIN_WRITE ∗ Method is a long-running or streaming operation : No. ∗ Filter for this method : protoPayload.methodName="google.firebase.database.v1beta.RealtimeDatabaseService.UndeleteDatabaseInstance" # Firebase Realtime Database audit logging ## Audit authentication information Audit log entries [include information](//cloud.google.com/logging/docs/reference/audit/auditlog/rest/Shared.Types/AuditLog#authenticationinfo)about the identity that performed the logged operation. To identify a request caller, see the following fields within the AuditLog object: ∗ Establishing realtime connections. Realtime Database Connect operations do not log authentication data since Realtime Database authenticates after a connection is established. Therefore, Connect has no authentication info. The AuthenticationInfo object contains a placeholder principalEmail of audit-pending-auth@firebasedatabase-{REGION_CODE}-prod.iam.gserviceaccount.com . ∗ Google Authentication. Realtime Database operations that use standard Google Authentication, such as traffic from Firebase Admin SDK or REST requests [authenticated with a standard OAuth token](/docs/database/rest/auth#google_oauth2_access_tokens), have an AuthenticationInfo object that contains the actual credentials email. ∗ Firebase Authentication . Realtime Database operations that use Firebase Authentication have an AuthenticationInfo object that contains a principalEmail value of audit-third-party-auth@firebasedatabase-{REGION_CODE}-prod.iam.gserviceaccount.com . The same is true if you implement your own authentication solution by minting custom JWTs. ∗ If a JSON Web Token (JWT) was used for third-party authentication, the thirdPartyPrincipal field includes the token's header and payload. For example, audit logs for requests authenticated with Firebase Authentication include that request's Firebase Authentication token. ∗ No authentication. Realtime Database operations that do not use any authentication have an AuthenticationInfo object that contains a principalEmail value of audit-no-auth@firebasedatabase-{REGION_CODE}-prod.iam.gserviceaccount.com A Realtime Database instance with open security rules may grant such requests. We recommend all users secure their databases properly. ∗ Legacy secrets tokens. Realtime Database operations using [legacy tokens](/docs/database/rest/auth#legacy_tokens)have an AuthenticationInfo object that contains a placeholder principalEmail of audit-secret-auth@firebasedatabase-{REGION_CODE}-prod.iam.gserviceaccount.com . For secrets-signed JWT, thirdPartyPrincipal contains the JWT headers and payload. ## Audit Firebase Security Rules evaluations Cloud Audit logs can be used to

identify requests that will be potentially affected by Security Rules changes. In the [AuthorizationInfo](//cloud.google.com/logging/docs/reference/audit/auditlog/rest/Shared.Types/AuditLog#authorizationinfo)object, authorization.permission can be one of: ∗ firebasedatabase.data.get : Read access granted at the path specified in resource . ∗ firebasedatabase.data.update : Write access granted at the path specified in resource . ∗ firebasedatabase.data.connect : Placeholder for Connect and Disconnect . No authorization required to connect to a Realtime Database instance. ∗ firebasedatabase.data.cancel : Used for Unlisten and OnDisconnectCancel . Revoking or canceling a previously-authorized operation requires no additional authorization. # Firebase Realtime Database audit logging ## Correlate Cloud Audit logs with Realtime Database profiler results You can perform in-depth performance analysis on Realtime Database using the [Realtime Database profiler](/docs/database/usage/profile)in combination with Realtime Database audit logging. Each tool has its strengths. | Cloud Audit Logging | Realtime Database profiler | |-|-| | ∗ Audits access to databases ∗ Continuously captures all requests ∗ Allows retrospective querying ∗ Contains detailed auth token info ∗ Incurs a [usage cost](//cloud.google.com/stackdriver/pricing#logging-costs) | ∗ Used for performance analysis ∗ Provides useful tooling for hotspot identification and thus performance optimization ∗ Can measure [listener-broadcast](/docs/cli/database-profile#listener-broadcast), which is not available in Audit logs due to potential data volume ∗ Lightweight and realtime, making it good for live load testing. Audit log entries may take a few minutes to appear. | Audit log contents correspond to profiler metrics as shown below. Note: Profiler operation types are discussed in this [reference guide](/docs/cli/database-profile). Metadata details for Realtime Database audit logs are discussed in the [reference documentation](/docs/reference/database/cal/rest/Shared.Types/RealtimeDatabaseAuditMetadata). | Audit Logging operation name | Special values in Realtime Database Audit Metadata | Profiler operation name | |-|-|-| | Connect | Request Type is REALTIME | concurrent-connect | | Disconnect | Request Type is REALTIME | concurrent-disconnect | | Read | Request Type is REALTIME | realtime-read | | Read | Request Type is REST | rest-read | | Write | Request Type is REALTIME | realtime-write | | Write | Request Type is REST | rest-write | | Update | Request Type is REALTIME . Check Precondition Type . | realtime-update realtime-transaction | | Update | Request Type is REST . Check Precondition Type . | rest-update rest-transaction | | ListenerListen | Request Type is REALTIME | listener-listen | | ListenerUnlisten | Request Type is REALTIME | listener-unlisten | | OnDisconnectPut | Request Type is REALTIME | on-disconnect-put | | OnDisconnectUpdate | Request Type is REALTIME | on-disconnect-update | | OnDisconnectCancel | Request Type is REALTIME | on-disconnect-cancel | | RunOnDisconnect | Request Type is REALTIME | run-on-disconnect |

- Audit logging for Firebase Cloud Messaging
  - # Audit logging for Firebase Cloud Messaging This document describes audit logging for Firebase Notifications Console. Google Cloud services generate audit logs that record administrative and access activities within your Google Cloud resources. For more

information about Cloud Audit Logs, see the following: ∗ [Types of audit logs](https://cloud.google.com/logging/docs/audit#types) ∗ [Audit log entry structure](https://cloud.google.com/logging/docs/audit#audit_log_entry_structure) ∗ [Storing and routing audit logs](https://cloud.google.com/logging/docs/audit#storing_and_routing_audit_logs) ∗ [Cloud Logging pricing summary](https://cloud.google.com/stackdriver/pricing#logs-pricing-summary) ∗ [Enable Data Access audit logs](https://cloud.google.com/logging/docs/audit/configure-data-access) ## Service name Firebase Notifications Console audit logs use the service name gcmcontextualcampaign-pa.googleapis.com . Filter for this service: protoPayload . serviceName = "gcmcontextualcampaign-pa.googleapis.com" ## Methods by permission type Each IAM permission has a type property, whose value is an enum that can be one of four values: ADMIN_READ , ADMIN_WRITE , DATA_READ , or DATA_WRITE . When you call a method, Firebase Notifications Console generates an audit log whose category is dependent on the type property of the permission required to perform the method. Methods that require an IAM permission with the type property value of DATA_READ , DATA_WRITE , or ADMIN_READ generate [Data Access](https://cloud.google.com/logging/docs/audit#data-access)audit logs. Methods that require an IAM permission with the type property value of ADMIN_WRITE generate [Admin Activity](https://cloud.google.com/logging/docs/audit#admin-activity)audit logs. API methods in the following list that are marked with (LRO) are long-running operations (LROs). These methods usually generate two audit log entries: one when the operation starts and another when it ends. For more information see [Audit logs for long-running operations](https://cloud.google.com/logging/docs/audit/understanding-audit-logs#lro). | Permission type | Methods | |-|-| | DATA_ READ | google. internal. gcm. contextual. campaign. v1. Campaign Service. Get Analytics google. internal. gcm. contextual. campaign. v1. Campaign Service. Get Audiences google. internal. gcm. contextual. campaign. v1. Campaign Service. Get Fcm Stats google. internal. gcm. contextual. campaign. v1. Campaign Service. Get Project Tracking Info google. internal. gcm. contextual. campaign. v1. Campaign Service. Get Topic List google. internal. gcm. contextual. campaign. v2. Firebase Notification Api Service. Get Analytics Label List google. internal. gcm. contextual. campaign. v2. Firebase Notification Api Service. List Campaigns | # Audit logging for Firebase Cloud Messaging ## API interface audit logs For information about how and which permissions are evaluated for each method, see the Cloud Identity and Access Management documentation for Firebase Notifications Console. ### google. internal. gcm. contextual. campaign. v1. Campaign Service The following audit logs are associated with methods belonging to google.internal.gcm.contextual.campaign.v1.CampaignService .Get Analytics ∗ Method : google.internal.gcm.contextual.campaign.v1.CampaignService.GetAnalytics ∗ Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) ∗ Permissions : ∗ firebasenotifications.messages.list - DATA_READ ∗ Method is a long-running or streaming operation : No. ∗ Filter for this method : protoPayload.methodName="google.internal.gcm.contextual.campaign.v1.

CampaignService.GetAnalytics" Get Audiences * Method : google.internal.gcm.contextual.campaign.v1.CampaignService.GetAudiences * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebasenotifications.messages.list - DATA_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.internal.gcm.contextual.campaign.v1.CampaignService.GetAudiences" Get Fcm Stats * Method : google.internal.gcm.contextual.campaign.v1.CampaignService.GetFcmStats * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebasenotifications.messages.list - DATA_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.internal.gcm.contextual.campaign.v1.CampaignService.GetFcmStats" Get Project Tracking Info * Method : google.internal.gcm.contextual.campaign.v1.CampaignService.GetProjectTrackingInfo * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebasenotifications.messages.list - DATA_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.internal.gcm.contextual.campaign.v1.CampaignService.GetProjectTrackingInfo" Get Topic List * Method : google.internal.gcm.contextual.campaign.v1.CampaignService.GetTopicList * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebasenotifications.messages.list - DATA_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.internal.gcm.contextual.campaign.v1.CampaignService.GetTopicList" ### google. internal. gcm. contextual. campaign. v2. Firebase Notification Api Service The following audit logs are associated with methods belonging to google.internal.gcm.contextual.campaign.v2.FirebaseNotificationApiService .Get Analytics Label List * Method : google.internal.gcm.contextual.campaign.v2.FirebaseNotificationApiService.GetAnalyticsLabelList * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebasenotifications.messages.list - DATA_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.internal.gcm.contextual.campaign.v2.FirebaseNotificationApiService.GetAnalyticsLabelList" List Campaigns * Method : google.internal.gcm.contextual.campaign.v2.FirebaseNotificationApiService.ListCampaigns * Audit log type : [Data access](https://cloud.google.com/logging/docs/audit#data-access) * Permissions : * firebasenotifications.messages.list - DATA_READ * Method is a long-running or streaming operation : No. * Filter for this method : protoPayload.methodName="google.internal.gcm.contextual.campaign.v2.FirebaseNotificationApiService.ListCampaigns"

- firebase.google.com/docs/ai-logic/cloud-audit-logging
  - # Audit logging for Firebase AI Logic This page describes the audit logs created by Firebase as part of [Cloud Audit Logs](https://cloud.google.com/logging/docs/audit/). ## Overview

Firebase services write audit logs to help you answer the questions, "Who did what, where, and when?". These are Cloud Audit Logs, provided as part of the [Google Cloud project connected to your Firebase project](//firebase.google.com/docs/projects/learn-more#firebase-cloud-relationship). Your Firebase projects each contain only the audit logs for resources that are directly within the project. For a general overview of Cloud Audit Logs, see [Cloud Audit Logs overview](https://cloud.google.com/logging/docs/audit/). For a deeper understanding of the audit log format, see [Understand audit logs](https://cloud.google.com/logging/docs/audit/understanding-audit-logs). ## Available audit logs The following types of audit logs are available for Firebase AI Logic: * Admin Activity audit logs Includes "admin write" operations that write metadata or configuration information. You can't disable Admin Activity audit logs. * Data Access audit logs Includes "admin read" operations that read metadata or configuration information. Also includes "data read" and "data write" operations that read or write user-provided data. To receive Data Access audit logs, you must explicitly enable them. For fuller descriptions of the audit log types, see [Types of audit logs](https://cloud.google.com/logging/docs/audit#types). ## Audited operations The following summarizes which API operations correspond to each audit log type in Firebase AI Logic: | Permission type | Methods | |-|-| | ADMIN_ READ | google. firebase. vertexai. v1beta. Config Service. Get Config | | ADMIN_ WRITE | google. firebase. vertexai. v1beta. Config Service. Update Config |
Message input

Gemini can make mista